

Luca Buiarelli

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Profile

Material Science PhD candidate with a physics background, specializing in first-principles theory of magnetic multipolar order, symmetry-constrained response functions, and unconventional magnetism in real materials, with experience connecting group-theoretical predictions to experimental observables.

Education

University of Minnesota, USA — PhD, Materials Science *Sep. 2023 – Present*

University of Copenhagen, Denmark — MSc, Physics *Sep. 2021 – May 2023*

University of Pisa, Italy — BSc, Physics *Sep. 2018 – Sep. 2021*

Articles

Published

1. Interplay between charge correlations and superconductivity across the superconducting domes of $\text{CsV}_3\text{Sb}_{5-x}\text{Sn}_x$

Andrea N. Capa Salinas, Brenden R. Ortiz, Steven J. Gomez Alvarado, Sarah Schwarz, Ganesh Pokharel, **Luca Buiarelli**, Hyeonseo Harry Park, Shiyu Yuan, Roland Yin, Suchismita Sarker, Turan Birol, Stephen D. Wilson

Phys. Rev. B **114** (2026) [[Journal](#)] [[arXiv](#)]

2. Probing multipolar order in the candidate altermagnet MnF_2 through the elastocaloric effect under strain

Rahel Ohlendorf, **Luca Buiarelli**, Hilary ML Noad, Andrew P Mackenzie, Rafael M Fernandes, Turan Birol, Jörg Schmalian, Elena Gati

Phys. Rev. Lett. **137** (2026) *Editors' Suggestion* [[Journal](#)] [[arXiv](#)]

3. Observation of mirror-odd and mirror-even spin texture in ultra-thin epitaxially-strained RuO_2 films
Yichen Zhang, Seung Gyo Jeong, **Luca Buiarelli**, Seungjun Lee, Yucheng Guo, Jiaqin Wen, Hang Li, Sreejith Nair, In Hyeok Choi, Zheng Ren, Ziqin Yue, Alexei Fedorov, Sung-Kwan Mo, Junichiro Kono, Jong Seok Lee, Tony Low, Turan Birol, Rafael M. Fernandes, Milan Radovic, Bharat Jalan, Ming Yi
Science Adv. **12** (2026) [[Journal](#)] [[arXiv](#)]

4. Electronic and structural properties of Rh- and Pd-based kagome layered shandites from first principles

Luca Buiarelli, Turan Birol, Brian M. Andersen, Morten H. Christensen

Phys. Rev. B **113** (2026) *Editors' Suggestion* [[Journal](#)] [[arXiv](#)]

5. Altermagnetic Polar Metallic Phase in Ultra-Thin Epitaxially-Strained RuO_2 Films

Seung Gyo Jeong, In Hyeok Choi, Sreejith Nair, **Luca Buiarelli**, Bitu Pourbahari, Jinyoung Oh, Nabil Bassim, Ambrose Seo, Woo Seok Choi, Rafael M. Fernandes, Turan Birol, Liuyan Zhao, Jong Seok Lee, Bharat Jalan

Proc. Natl. Acad. Sci. **123** (2026) [[Journal](#)] [[arXiv](#)]

6. Noncollinear Magnetic Multipoles in Collinear Altermagnets

Luca Buiarelli, Rafael M. Fernandes, Turan Birol

Phys. Rev. B **112**, 224442 (2025) [[Journal](#)] [[arXiv](#)]

Preprints

1. Piezomagnetism in a model cubic noncollinear altermagnet

Sudarshan Sharma, **Luca Buiarelli**, Richard Spieker, Ivan Jakovac, Damjan Pelc, Turan Birol, Martin Greven

Submitted [[arXiv](#)]

2. Discovery of an odd-parity f-wave charge order in a kagome metal

Jiangchang Zheng, Caiyun Chen, Ruiqin Fu, **Luca Buiarelli**, Zihan Lin, Fazhi Yang, Tianhao Guo,

Ganesh Pokharel, Andrea Capa Salinas, Sen Zhou, Turan Birol, Stephen D. Wilson, Junzhang Ma, Daniel J. Schultz, Xianxin Wu, Berthold Jäck
Submitted [[arXiv](#)]

Workshops and Conferences

- Aspen Center for Physics, Winter Conference on Altermagnetism, Aspen *April 2026*
“Altermagnetic piezomagnetism and spin-lattice coupling” ([poster](#))
- American Physical Society March Meeting, Denver *March 2026*
“ProDenCeR: a python code to project first-principles densities onto harmonics and representations” ([contributed talk](#))
- Flatiron Institute, TRIQS School, Paris *September 2025*
- American Physical Society March Meeting, Los Angeles *March 2025*
“Magnetic multipoles in altermagnets” ([contributed talk](#))
- School on Electron–Phonon Physics, University of Texas at Austin *June 2024*
- American Physical Society March Meeting, Minneapolis *March 2024*
“Structural properties of kagome-layered crystals” (contributed talk)

Awards and Fellowships

Matthew Tirrell Graduate Fellowship *2024*

Teaching

Teaching Assistant *Spring and Fall 2024*
University of Minnesota, Introduction to Materials Science Laboratory (MATS 2001/2002)

Service and Outreach

Journal Refereeing
Nature *npj Quantum Materials*, Nature *Communications Materials*

Penne Amiche della Scienza *2025-Present*
Science outreach program for Italian middle school students involving monthly letter exchanges, aimed at fostering interest in science and academic careers.

Skills

Coding: Python, Matlab, C/C++

First-principles: Abinit, VASP, Wannier90, pythtb

Computing: Experience with HPC clusters, SLURM job scheduling, and MPI-parallelized workflows for large-scale DFT calculations.

Scientific Software Development: Developed multiple post-processing tools for DFT and Wannier tight-binding Hamiltonians, including [ProDenCeR](#), a tool to project first-principles densities onto harmonics and group representations, and an independent implementation of Wannier-based nonlinear optical conductivity interpolation.

Languages: English (fluent), Italian (native), Spanish (basic)